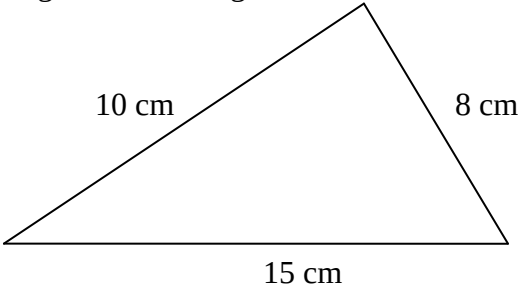
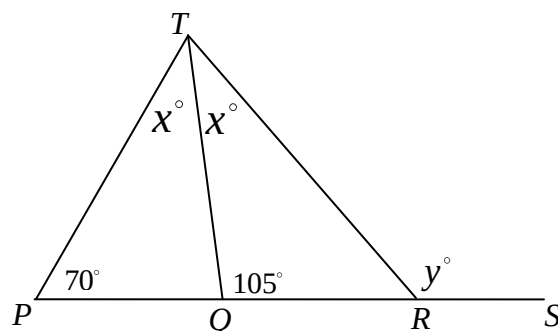


|                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                        |                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                      | <p style="text-align: center;"><b>BAULKHAM HILLS HIGH SCHOOL</b><br/> <b>YEAR 11 MATHEMATICS TASK 2</b><br/> <b>JUNE 2009</b></p> <p><i>Time Allowed: 70 minutes</i></p>                                                                                                               |                                                                                                                        |
| <p>Instructions:</p>                                                                                                                                                                                                                  | <p>a) Write all your answers on your own paper.<br/> b) Show all necessary working.<br/> c) Marks may be deducted for careless or badly arranged work.<br/> d) No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.</p> |                                                                                                                        |
| <p>1. Find the exact value of;</p> <p>a) <math>\tan 330^\circ</math></p> <p>b) <math>\sec 120^\circ</math></p>                                                                                                                        |                                                                                                                                                                                                                                                                                        | <p style="text-align: right;"><b>Marks</b></p> <p style="text-align: right;">1</p> <p style="text-align: right;">1</p> |
| <p>2. If <math>\tan \theta = \frac{3}{5}</math> and <math>\theta</math> is a reflex angle, find the exact value of <math>\cos \theta</math></p>                                                                                       |                                                                                                                                                                                                                                                                                        | <p style="text-align: right;">2</p>                                                                                    |
| <p>3. A ship leaves port and sails on a bearing of <math>120^\circ \text{T}</math> for 50 km.</p> <p>a) Draw a sketch to show this information.</p> <p>b) How far east of the port is it at this time, correct to the nearest km?</p> |                                                                                                                                                                                                                                                                                        | <p style="text-align: right;">1</p> <p style="text-align: right;">2</p>                                                |
| <p>4. Sketch <math>y = \tan x</math> for <math>0^\circ \leq x \leq 360^\circ</math></p>                                                                                                                                               |                                                                                                                                                                                                                                                                                        | <p style="text-align: right;">2</p>                                                                                    |
| <p>5. Find the largest angle in this triangle, correct to the nearest minute.</p> <div style="text-align: center;">  </div>                        |                                                                                                                                                                                                                                                                                        | <p style="text-align: right;">3</p>                                                                                    |
| <p>6. Prove that the points <math>A(1,2)</math>, <math>B(-1,6)</math> and <math>C(2,0)</math> are collinear.</p>                                                                                                                      |                                                                                                                                                                                                                                                                                        | <p style="text-align: right;">3</p>                                                                                    |
| <p>7. If <math>(2,3)</math> is the midpoint of the two points <math>A(x,y)</math> and <math>B(4,4)</math>, find the coordinates of <math>A</math>.</p>                                                                                |                                                                                                                                                                                                                                                                                        | <p style="text-align: right;">3</p>                                                                                    |
| <p>8. A line is represented by the equation <math>x + 2y - 6 = 0</math>. Find the equation of the line that is perpendicular to this line and passes through the point <math>(1,3)</math>.</p>                                        |                                                                                                                                                                                                                                                                                        | <p style="text-align: right;">3</p>                                                                                    |

**Marks**

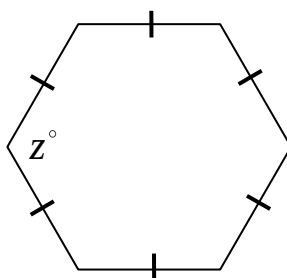
9. Find the value of each pronumeral, (no reasons required);

a)



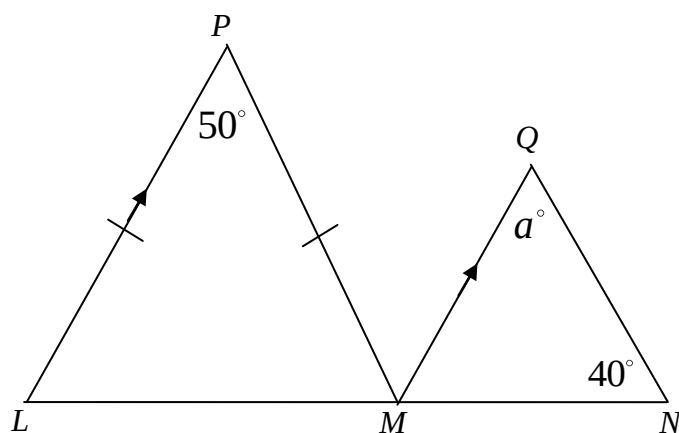
2

b)



1

c)



2

10. Solve for  $0^\circ \leq \theta \leq 360^\circ$

a)  $4\cos^2 \theta = 3$

3

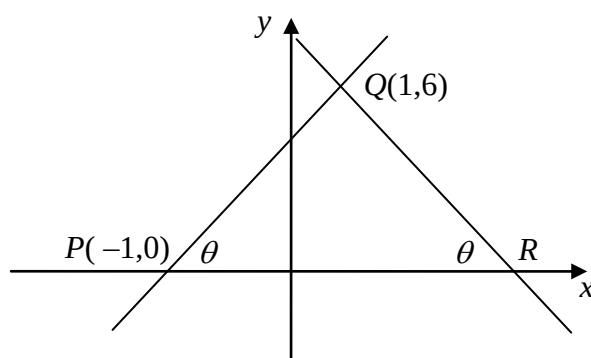
b)  $2\sin \theta - 4\cos \theta = 0$

3

c)  $\tan 2\theta = \frac{1}{\sqrt{3}}$

3

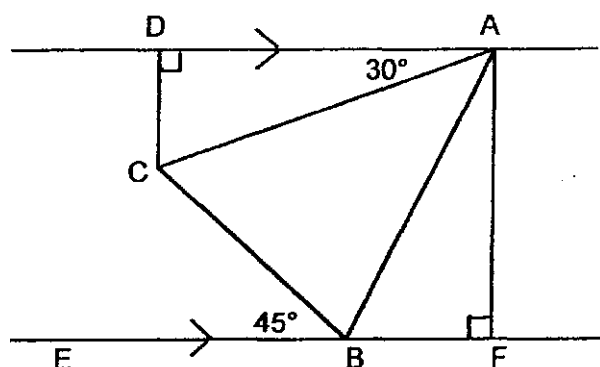
11.



In the diagram,  $P$  and  $Q$  have coordinates  $(-1, 0)$  and  $(1, 6)$  and  $\angle QPR = \angle QRP = \theta$

- |                                                           |   |
|-----------------------------------------------------------|---|
| a) Find the length of $PQ$ as a surd.                     | 2 |
| b) Show that $\tan \theta = 3$                            | 2 |
| c) Show that the gradient of $QR$ is $-3$ . Give reasons. | 2 |
| d) Show that the equation of $QR$ is $3x + y - 9 = 0$ .   | 2 |
| e) Find the coordinates of $R$ .                          | 1 |
| f) Find the perpendicular distance from $P$ to $QR$ .     | 2 |
| g) Hence calculate the area of $\triangle PQR$            | 2 |

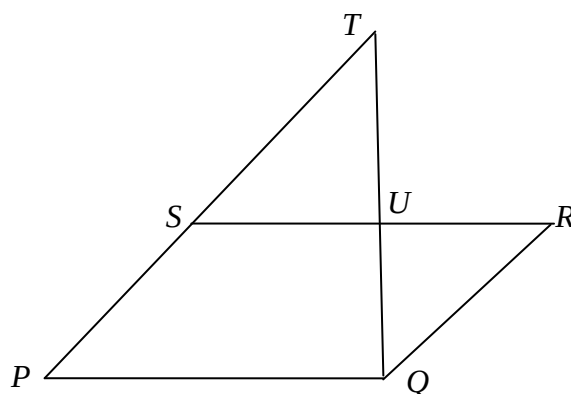
12.



In the diagram above  $AC = AB$  and  $DA \parallel EF$ .  $\angle DAC = 30^\circ$  and  $\angle CBE = 45^\circ$ .  $CD \perp DA$  and  $DA \perp AF$ .

- |                                                         |   |
|---------------------------------------------------------|---|
| a) Find the size of $\angle ACB$ , giving reasons.      | 3 |
| b) Hence find the size of $\angle CAB$ , giving reasons | 2 |
| c) Prove that $\triangle ACD \equiv \triangle ABF$      | 3 |

13.



$PQRS$  is a parallelogram and  $PT$  is a straight line through  $S$ .  $TQ \perp PQ$

a) Prove  $\triangle URQ \parallel \triangle QPT$

3

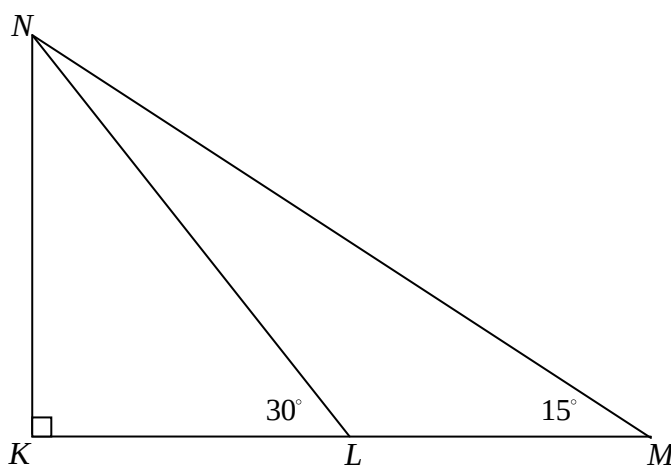
b) If  $QR = 5$ ,  $UR = 4$ ,  $ST = 8$ , find  $TP$  and  $SU$  and hence  $TQ$ .

5

14. Prove that  $\frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\operatorname{cosec} \theta + \cot \theta} = 2 \cot \theta$

2

15.



a) If  $LM = 1$  metre, explain why  $LN$  also equals 1 metre.

2

b) Use the sketch to deduce that  $\tan 15^\circ = 2 - \sqrt{3}$

3

# Year 11 Mathematics Task 3 Solutions / 471

1/a)  $\tan 330^\circ = -\tan 30^\circ$   
 $= -\frac{1}{\sqrt{3}}$  ①

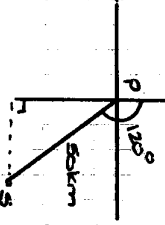
6/  $m_{AB} = \frac{6-2}{-1-1} = \frac{4}{-2} = -2$   
 $m_{BC} = \frac{0-6}{2+1} = \frac{-6}{3} = -2$   
 $\therefore m_{AB} = m_{BC}$   
 $\therefore A, B, C$  are collinear ③

b)  $\sec 120^\circ = \frac{1}{\cos 120^\circ}$   
 $= \frac{1}{-\frac{1}{2}} = -2$  ①

7/  $\frac{x+4}{2} = 2$   
 $x+4 = 4$   
 $x = 0$   
 $y+4 = 6$   
 $y = 2$   
 $\therefore A$  is the point  $(0, 2)$  ③

2/  $\tan \theta = \frac{3}{4}$ ,  $\theta = \tan^{-1} \frac{3}{4}$   
 $\therefore 103.4^\circ < 270^\circ$   
 $\cos \theta = -\frac{4}{5}$  ②

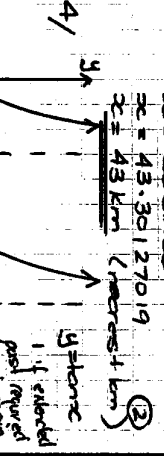
8/  $x+2y-6 = 0 \Rightarrow m = -\frac{1}{2}$   
 $\therefore$  required  $m = 2$  ①

3/a)  ①

9/a)  $x+70 = 105$   
 $x = 35$  ①

b)  $\frac{x}{50} = \sin 60^\circ$   
 $x = 50 \sin 60^\circ$   
 $x = 43.30127019$  ③

10/a)  $4 \cos^2 \theta = 3$   
 $\cos^2 \theta = \frac{3}{4}$   
 $\cos \theta = \pm \frac{\sqrt{3}}{2}$   
 $\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$  ③

4/  ①

11/a)  $RQ = \sqrt{6^2 + 2^2} = \sqrt{40}$   
 $= 2\sqrt{10}$  units ②

5/ largest angle is opposite the longest side.  
 $\cos \theta = \frac{8^2 + 10^2 - 15^2}{2 \times 8 \times 10} = -1$   
 $\theta = 112.25^\circ$  ③

b)  $6z = 180(6-z)$   
 $z = 120$  ①

c)  $2 \angle PML + 50 = 180$   
 $2 \angle PML = 130$   
 $\angle PML = 65^\circ$   
 $\angle AMQ = 50^\circ$   
 $a + 40 = 65 + 50$   
 $a = 75^\circ$  ③

12/a)  $\tan 2\theta = \frac{1}{\sqrt{3}}$   
 $2\theta = 30^\circ$   
 $\theta = 15^\circ$   
 $\therefore \theta = 15^\circ, 105^\circ, 195^\circ, 285^\circ$  ③

13/  $PT \parallel QR$  (opposite sides in  $\triangle PQR$ )  
 $\angle PQR = \angle RQR$  (alternate  $\angle$ s,  $PT \parallel QR$ )  
 $\angle PQR = \angle RQR$  (vertical  $\angle$ s in  $\triangle PQR$ )  
 $\therefore \triangle PQR \cong \triangle RQR$  (AAS) ③

14/a)  $\tan \theta = \frac{3}{4}$   
 $\theta = \tan^{-1} \frac{3}{4}$   
 $\therefore \theta = 36.87^\circ$   
 $\therefore \theta = 36.87^\circ, 143.13^\circ, 216.87^\circ, 323.13^\circ$  ③

15/  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

16/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

17/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

18/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

19/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

20/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

21/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

22/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

23/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

24/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

25/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

26/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

27/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

28/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

29/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

30/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

31/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

32/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

33/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

34/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

35/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

36/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

37/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

38/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

39/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

40/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

41/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

42/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

43/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

44/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

45/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

46/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

47/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

48/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

49/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

50/a)  $\triangle ABC$  is isosceles  
 $\therefore QR = 2\sqrt{10}$  ①

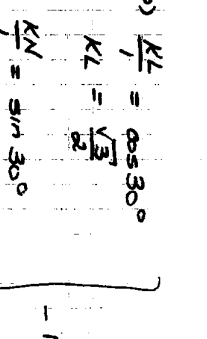


Figure 1. Schematic representation of the experimental design. The subjects were divided into two groups: the control group and the experimental group. The control group was divided into two subgroups: the control group and the control group. The experimental group was divided into two subgroups: the experimental group and the experimental group. The control group was divided into two subgroups: the control group and the control group. The experimental group was divided into two subgroups: the experimental group and the experimental group.

2

$$\tan 15^\circ = \frac{1}{\sqrt{3}}$$

$\frac{KL}{K_L + K}$

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+

$$\sqrt{2+2}$$

18

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81

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Figure 1 is a line graph illustrating the percentage of the total sample for various age groups over time. The x-axis represents years from 1950 to 2000, and the y-axis represents the percentage of the total sample, ranging from 0 to 100. The age groups are: 0-14, 15-24, 25-34, 35-44, 45-54, 55-64, 65-74, and 75+. The graph shows a general trend of decreasing percentages for younger age groups and increasing percentages for older age groups over time.

| Year | 0-14 | 15-24 | 25-34 | 35-44 | 45-54 | 55-64 | 65-74 | 75+ |
|------|------|-------|-------|-------|-------|-------|-------|-----|
| 1950 | 25   | 20    | 15    | 10    | 5     | 2     | 1     | 0   |
| 1960 | 22   | 18    | 14    | 11    | 6     | 3     | 2     | 1   |
| 1970 | 18   | 15    | 12    | 13    | 8     | 4     | 3     | 2   |
| 1980 | 15   | 12    | 10    | 15    | 10    | 6     | 5     | 3   |
| 1990 | 12   | 10    | 8     | 18    | 12    | 8     | 7     | 4   |
| 2000 | 10   | 8     | 6     | 20    | 14    | 10    | 9     | 5   |

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